Yixuan Yang

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Duke UNIVERSITY

Education

Duke University *08/2024 – Present* *Durham, United States*
Ph.D. in Electrical and Computer Engineering, Advisor: ***Prof. Rishi Kamaleswaran*** GPA: 3.96/4.0

Southern University of Science and Technology (SUSTech) *08/2020 – 06/2024* *Shenzhen, China*
B.E. in Computer Science, Academic Advisor: ***Prof. Xin Yao*** GPA: 3.53/4.0

- Distinguished Graduate, Department of Computer Science (Top 1%, 2 out of 245 students)
- Guo Xie Bi Rong Fellowship (Top 4%); SUSTech Outstanding Undergraduate Alumnus, Class of 2024

University of California, Berkeley (UCB) *08/2022 – 01/2023* *Berkeley, United States*
Exchange Undergraduate Student GPA: 4.0/4.0

Research Interests

World Models & Latent Dynamics, Self-Supervised Representation Learning, Sequential Decision-Making under Uncertainty, Embodied & Clinical Foundation Models.

Research Experiences

Clin-JEPA: Joint-Embedding Predictive Pretraining on EHR Patient Trajectories *08/2025 – Present*
Duke University – Kamaleswaran Lab ***Advised by Prof. Rishi Kamaleswaran***

- Proposed *Clin-JEPA*, the first JEPA to co-train an LLM encoder against the rollout of a **retained** latent predictor; prior designs discard the predictor or train it on a frozen encoder. Co-training spreads the representation over **55% more effective dimensions**, and the same encoder paired with a conventionally trained predictor loses long-horizon accuracy entirely.
- Encoding patients as **clinical text** rather than fixed-vocabulary codes lets any observation reach the encoder with **no imputation, normalization, or feature engineering**; one backbone serves **15 tasks** from one embedding set, outperforming **CLMBR-T-base**, pretrained on **40× more patients**.

Contact-Rich Humanoid Manipulation for Scientific Laboratory Automation *08/2024 – 03/2025*
Duke University – General Robotics Lab ***Advised by Prof. Boyuan Chen***

- Engineered a manipulation learning platform for the Unitree G1 humanoid in Nvidia Isaac Gym, pairing a physical instrument testbed with a dimensionally matched simulation counterpart.
- Designed the full task stack for contact-rich manipulation of laboratory instrument controls, from the inverse-kinematics control interface and fingertip contact sensing to the reward and termination structure trained with PPO.

SniffySquad: Patchiness-Aware Gas Source Localization by Collaborative Robots *05/2023 – 06/2024*
Tsinghua University (Visiting Research) ***Advised by Prof. Xinlei Chen***

- Proposed *SniffySquad*, recasting gas-source search as MCMC sampling: each robot runs a **Langevin diffusion** over a shared belief map, its **temperature setting its role**, with robots exchanging roles by a **replica-exchange** acceptance rule.
- Achieved 20%+ higher success rate and 30%+ path efficiency over SOTA baselines in simulation and a real-robot testbed.

Machine Learning-Based Problem Reduction for Large-Scale UFLP Optimization *01/2023 – 06/2024*
Southern University of Science and Technology ***Advised by Prof. Xin Yao***

- Proposed a learning-based reduction framework for UFLP: costs become ranks **normalized by instance size**, so four distribution statistics describe a facility identically at any scale, letting a small-instance classifier transfer to large ones.
- Pruned instances to **13.2%** of their facilities where only **11.6% are essential**, accelerating four benchmark meta-heuristics.

Publications

- **Yixuan Yang**, Mehak Arora, Ryan Zhang, Baraa Abed, Junseob Kim, Tilendra Choudhary, Md Hassanuzzaman, Kevin Zhu, Ayman Ali, Chengkun Yang, Alasdair Edward Gent, Victor Moas, and Rishikesan Kamaleswaran. **Clin-JEPA: A Multi-Phase Co-Training Framework for Joint-Embedding Predictive Pretraining on EHR Patient Trajectories**. ***arXiv preprint arXiv:2605.10840***, May 2026.
- Yuhan Cheng*, Xuecheng Chen*, **Yixuan Yang**, Haoyang Wang, Jingao Xu, Chaopeng Hong, Susu Xu, Xiao-Ping Zhang, Yunhao Liu, and Xinlei Chen. **SniffySquad: Patchiness-Aware Gas Source Localization with Multi-Robot Collaboration**. ***ACM Transactions on Sensor Networks, Feb 2026***. DOI: 10.1145/3786599.
- Yuhan Cheng*, Xuecheng Chen*, **Yixuan Yang**, Haoyang Wang, Yuxuan Liu, and Xinlei Chen. **Poster: Olfactory Sensing in Turbulent Airflow via Collaborative Robots**. In ***Proceedings of the 25th International Workshop on Mobile Computing Systems and Applications (ACM HotMobile)***, 2024.
- Shuaixiang Zhang, **Yixuan Yang**, Hao Tong, and Xin Yao. **Learning-Based Problem Reduction for Large-Scale Uncapacitated Facility Location Problems**. In ***IEEE Congress on Evolutionary Computation (CEC)***, 2024.

Academic Service & Technical Skills

Government Service: Student Researcher, **North Carolina AI Leadership Council**, **Tech, Data, & Infra team** (01/2026 – 06/2026); work acknowledged in the official **NC Statewide AI Strategic Roadmap** (announced by Gov. Josh Stein, Jul 2026).
Teaching Assistant: ECE 590: Cross-Platform Mobile App Programming, Duke ECE (Fall 2026); DECISION 520Q/618: Data Science for Business, Duke Fuqua School of Business (Fall 2024).
Reviewer: ACL Rolling Review (ARR, 2026); ICLR 2026 Workshop on Foundation Models for Science (FM4Science, 2026); IEEE International Conference on Healthcare Informatics (ICHI, 2026); IEEE Transactions on Circuits and Systems for Video Technology (TCSVT, 2026); IEEE Transactions on Multimedia (TMM, 2026); IEEE J-STSP (Intelligent Robotics SI, 2023).
Tools: NVIDIA Isaac Gym & Genesis, hardware prototyping (Raspberry Pi, Arduino, Fusion 360, UWB systems); agentic AI coding (Claude Code, OpenAI Codex); MIMIC-IV / EHR databases (credentialed access).